

POL DEL CASTILLO

M.S. in Mechanical Engineering candidate @ RPI · John Wen Research Group @ RPI

✉ polca2003@gmail.com · 📞 (912) 541-2635 · 🏠 Troy, NY · 🌐 pol-del-castillo · 📧 pd05849

- Former Division 1 student-athlete as a member of the Georgia Southern men's tennis team with strong work ethic, discipline, and teamwork
- Passionate about using robotics and mechanical design for innovative medical devices, including robotic arms, surgical robots, and prosthetics
- Experienced in SolidWorks, AutoCAD, Rapid prototyping (3D printing), Python, MATLAB, Julia, GitHub, ML

EDUCATION

Rensselaer Polytechnic Institute

August 2025 – Expected Graduation May 2027
Troy, NY

M.S. in Mechanical Engineering, Department of Mechanical, Aeronautical and Nuclear Engineering
Research focus: Robotics and manufacturing, with interests in applications to medical devices, robotics, and advanced manufacturing
GPA: **3.77**

Georgia Southern University

August 2021 – May 2025
Statesboro, GA

B.S. in Mechanical Engineering, Allen E. Paulson College of Engineering and Computing
GPA: **3.79/4.00**; *Magna Cum Laude*; Dean's List; Student-Athlete of the Year 2023-2024

GRADUATE RESEARCH

Universal Manipulation Interface (UMI) integration

August 2025 – Present
Troy, NY

Center for Smart Convergent Manufacturing Systems

- Implemented the Universal Manipulation Interface (UMI) to accelerate the collection of data for **robot** training by using a low-cost gripper
- Studied **diffusion policies**, a form of **generative AI**, to create diverse and robust robot trajectories from human demonstrations

SO-101 LeRobot Arm Project

August 2025 – Present
Troy, NY

Center for Smart Convergent Manufacturing Systems

- 3D-printed and assembled the LeRobot SO-101 **robotic arm** to work on **machine learning** models and collected data through teleoperation to train multiple of them, like ACT or diffusion policy
- Implemented IK-Geo, an **inverse kinematics** solver for 6-DOF revolute robots, faster, more accurate, and stable than other solvers like IKFast

UNDERGRADUATE RESEARCH

Lithium-ion Battery Degradation Modelling

August 2025 – April 2025
Statesboro, GA

Fire and Energy Laboratory Research

- Utilized the **SciML** ecosystem in Julia to develop a hybrid **neural network** model that integrates physics-based battery models
- The neural networks were able to learn complex degradation mechanisms directly from data

WildFire Spread Simulator

January 2024 – May 2025
Statesboro, GA

Fire and Energy Laboratory Research

- Developed an immersed boundary method for drag coefficient calculations in complex rigid and flexible vegetation for a research symposium
- Used a Navier-Stokes equation solver in Julia to numerically **simulate** wind flows through complex rigid and flexible vegetation

PROJECTS

Automation of Linear Scale System

January 2025 – April 2025
Savannah, GA

Howmet Aerospace

- Led a project to modernize an existing **bandsaw** system at Howmet Aerospace, implementing a machine automation system for precise billet measurement and cutting operations. This **retrofit** improved efficiency and precision without a full equipment replacement
- Designed and developed a fully functional working **prototype** featuring a programmable FUYU FSL80 linear guide with a stepper motor. This system automatically measures billet lengths to streamline the cutting process

LEADERSHIP

Student Athlete, Team Captain

August 2021 – May 2025
Statesboro, GA

Georgia Southern Men's Tennis Team

- Successfully managed a **full-time** academic schedule while committing to over **20 hours per week** of training as a Division I student-athlete
- Led the team as **captain** to a year-end national ranking (No. 66 out of 264) and a Sun Belt Regular Season Championship in 2024
- Named the 2023-2024 Georgia Southern University **Student-Athlete of the Year**